COURSE SLIDES · NON-WSQ

Git and Github Masterclass



Course Code: C171

Exam: GH-900 · 100 minutes
Version v3.0 · 4 October 2026

Tertiary Infotech Academy Pte Ltd · UEN 201200696W

COURSE ADMINISTRATION

Welcome & Housekeeping

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About the Trainer

GENERAL TRAINER TEMPLATE — THE DELIVERING TRAINER COMPLETES THIS PROFILE BEFORE CLASS



Trainer Name

Job Title · Organisation

ROLE

Current role and years of relevant industry experience

FOCUS

Primary technical domains and specialisations

DELIVERY

Certifications held and courses delivered

APPROACH

Training philosophy and how learners are supported

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COURSE ADMINISTRATION

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

ROLE

Founder and principal trainer

FOCUS

Cloud, DevOps, automation and AI-enabled development

DELIVERY

Certification-aligned technical training

APPROACH

Concept clarity, hands-on evidence and exam readiness

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HOUSEKEEPING

Ground Rules

- Participate actively.
- One conversation at a time.
- Ask questions early.
- Respect different experience levels.
- Return from breaks on time.
- Protect credentials and never share secrets.

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VISUAL COURSE PORTAL GUIDE

Download Course Material

1. Open lms-tms.tertiaryinfotech.com
2. Sign in with your registered email
3. Open this course registration
4. Download slides, Learner Guide and Labs

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7.5 HOURS · 14 LABS

One-Day Lesson Plan

Morning · Git by doing

01 Install Git, Git Bash and Desktop

02 Stage, commit and read history

03 Branch, merge and synchronise

04 Resolve conflicts and recover safely

Afternoon · Collaborate and govern

01 PRs, issues and review

02 Actions, Codespaces and Copilot

03 Projects, security and community

04 GH-900 evidence-based preparation

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EXAM BLUEPRINT

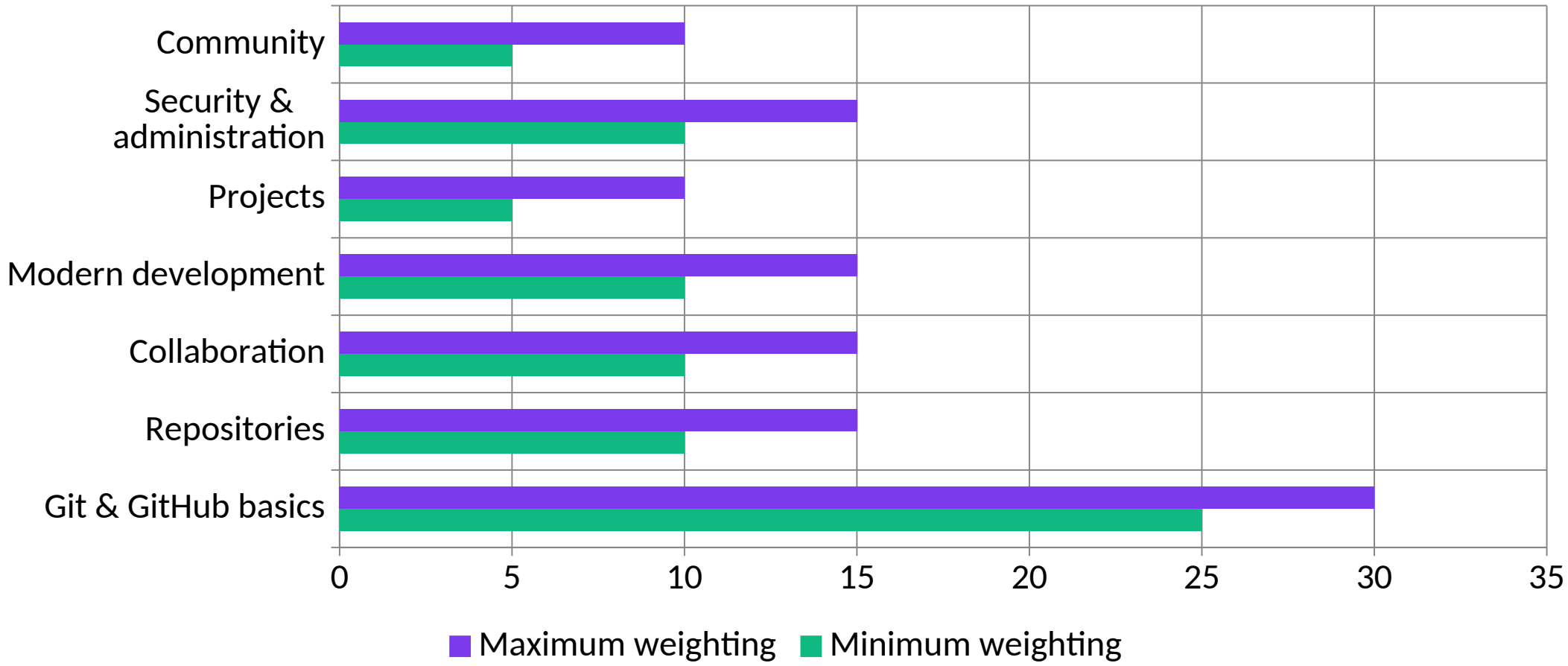
GitHub Foundations (GH-900)

Seven domains · beginner level · proctored through Pearson VUE

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EDITABLE NATIVE POWERPOINT CHART

GH-900 Domain Weightings



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01

DOMAIN 01 · 25–30%

Understand Git and GitHub Basics

Version control · Git areas and states · commands · accounts · GitHub Flow · Markdown

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EXAM WEIGHTING 25–30%

Key Concepts — Understand Git and GitHub Basics

Version control

History, traceability, rollback and safe parallel work.

Git vs GitHub

Distributed version control locally; collaboration platform in the cloud.

Core objects

Repository, working tree, staging area, commit, branch, HEAD, remote and tag.

GitHub Flow

Branch → commit → pull request → review → merge → deploy.

Account choices

Personal accounts, organisations and enterprises solve different governance needs.

Command literacy

status, diff, add, commit and log turn file changes into deliberate snapshots.

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FOUNDATIONAL DISTINCTION

Git and GitHub Are Complementary

Git · local DVCS

01 Tracks snapshots and history

02 Branches cheaply

03 Works offline

04 Pushes and pulls through remotes

GitHub · collaboration platform

01 Hosts Git repositories

02 Adds issues, PRs and review

03 Adds automation and security

04 Connects a global community

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PREPARE ONCE · VERIFY BEFORE LABS

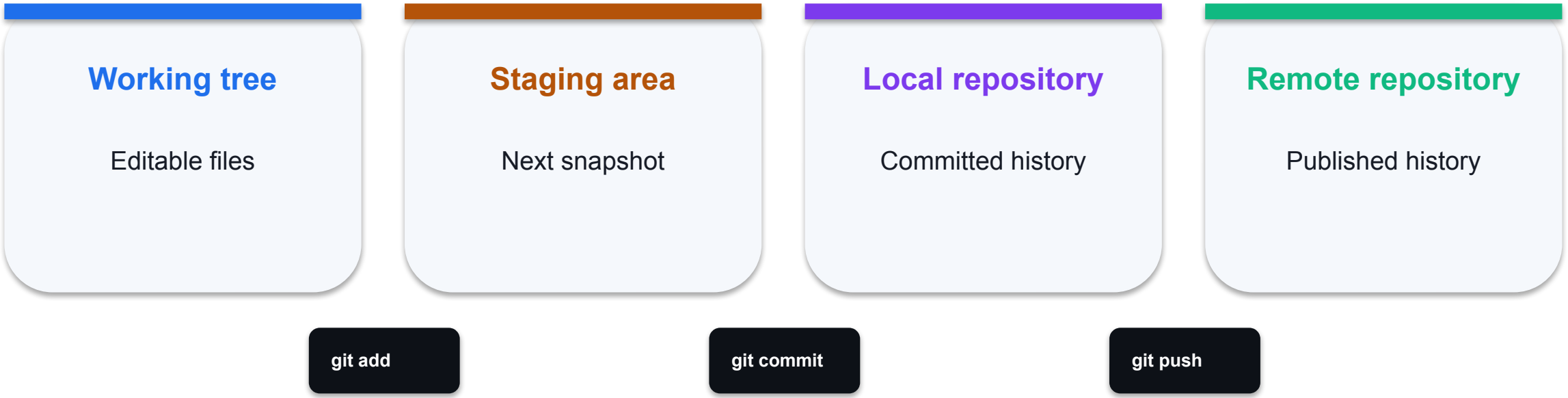
Toolchain Setup

1. Install Git
2. Open Git Bash or terminal
3. Configure identity
4. Install GitHub Desktop
5. Sign in and verify

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FILES MOVE; COMMANDS CHANGE THE RECORDED STATE

The Four Git Areas



git fetch updates remote-tracking references; git pull also integrates into the current branch.

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STATUS REPORTS MULTIPLE DIMENSIONS

File Conditions Are Not One Linear State

Untracked

A working-tree file Git does not yet track.

Modified

A tracked working-tree file differs from HEAD.

Staged

Selected content differs between the index and HEAD.

Committed

The snapshot is recorded in local history.

Remote branch updated

Push publishes commits; this is not a file state.

Key insight

One file can have staged content plus newer unstaged edits.

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OBSERVE · DECIDE · ACT · VERIFY

Inspect Before You Change State

git status -sb

Where am I and what changed?

git diff

What is unstaged?

git diff --staged

What will the next commit contain?

git log --oneline

What history already exists?

git remote -v

Where will fetch and push connect?

git branch -vv

What does this branch track?

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COMMIT ANATOMY

A Commit Is More Than a Save

Tree

Snapshot of staged project content.

Parent

Link to the previous commit or commits.

Identity

Author and committer metadata.

Time

When the snapshot was recorded.

Message

Why the outcome matters.

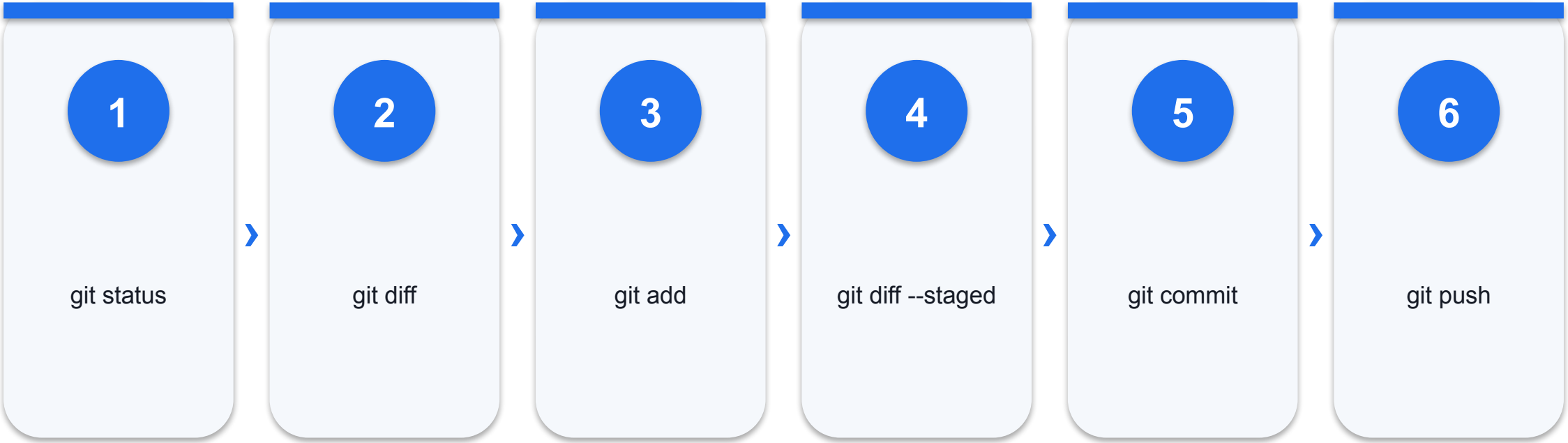
Hash

Content-derived object identifier.

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INSPECT BEFORE AND AFTER

Safe Daily Command Loop



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CHOOSE THE INTERFACE; KEEP THE MENTAL MODEL

GitHub Desktop and CLI Say the Same Thing

GitHub Desktop

01 Changes

02 Commit to branch

03 Current branch

04 Fetch / Pull / Push

05 History

Git commands

01 status + diff

02 commit

03 branch + switch

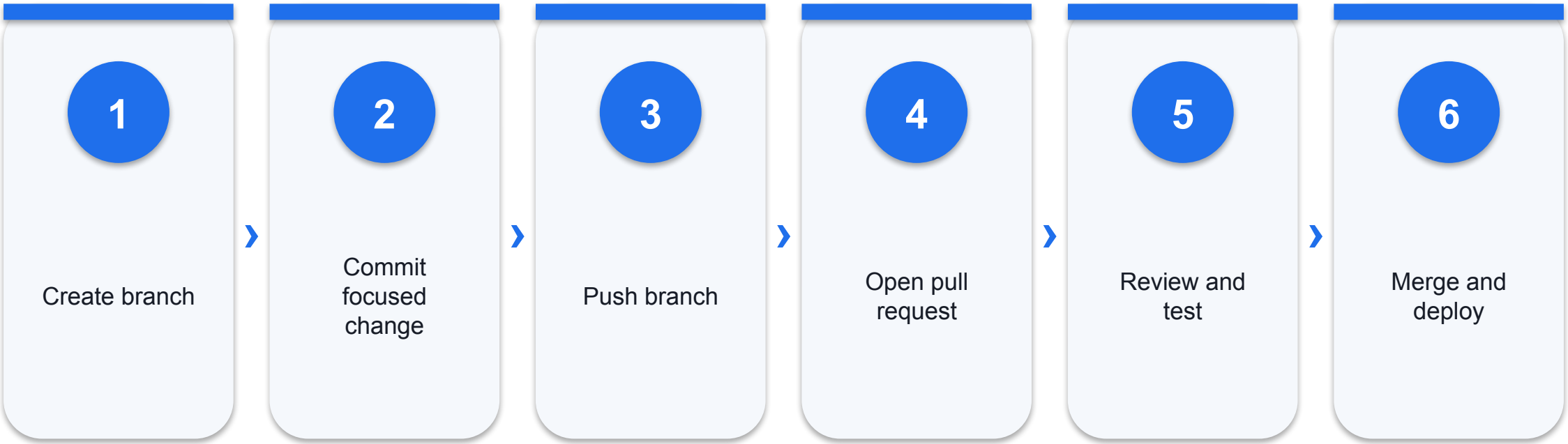
04 fetch / pull / push

05 log + show

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COLLABORATION LOOP

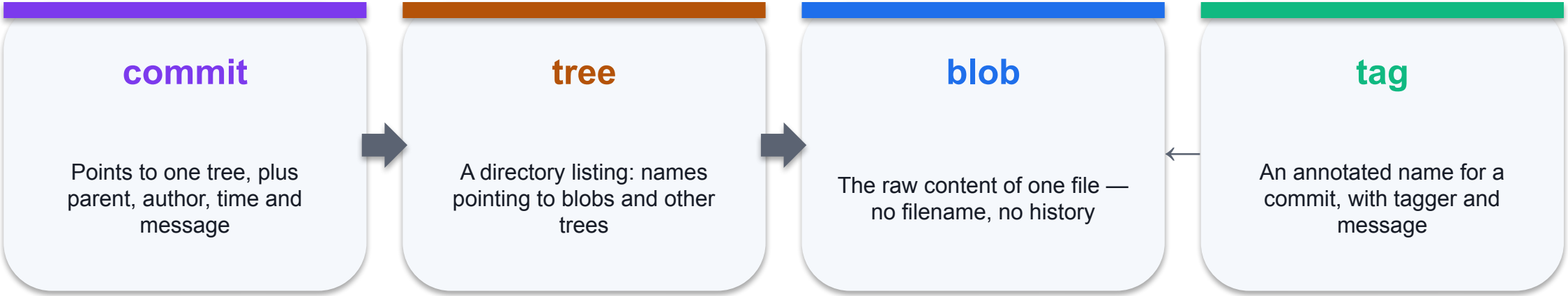
GitHub Flow



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CONTENT-ADDRESSED BY SHA-1 HASH

Git Stores Four Kinds of Object




```
git cat-file -p HEAD
```

```
git hash-object -w <file>
```

Every object is named by the SHA-1 hash of its own content.

Identical content stores once, whatever the filename. Change one byte and the blob hash changes, so the tree hash changes, so the commit hash changes — which is why history cannot be edited silently and why a commit ID is proof of exact content.

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ACCOUNTS ARE NOT PLANS

Three Account Types Solve Three Problems

Personal and organisation

01 Personal: your identity, owns repositories and acts under your name

02 Organisation: shared ownership, unlimited collaborators, team permissions

03 A personal account can belong to many organisations

04 Owners and security managers hold administrative control

Enterprise

01 Groups many organisations under one handle

02 Central policy, billing and security enforcement

03 Enables InnerSource across organisations

04 Cloud or Server deployment; EMU provisions users from the IdP

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PLAN SELECTION IS A GOVERNANCE DECISION

GitHub Plans and What They Add

Free (personal)

Unlimited public and private repos, Dependabot alerts, 2FA, 2,000 Actions minutes/month, 500 MB Packages.

Free (organisation)

Everything above plus team access controls; private repos have limited features.

Pro

Personal plan adding 3,000 Actions minutes, protected branches, code owners, Pages, Wikis and repository insights on private repos.

Team

Pro capability at organisation level plus draft PRs, team reviewers, scheduled reminders and Codespaces control.

Enterprise

SAML SSO, SCIM provisioning, GitHub Connect, Advanced Security option, 99.9% SLA and centralised policy.

Billing model

Each account is billed separately: fixed subscriptions plus usage-based charges for Actions minutes, storage and Copilot seats.

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SAME PLATFORM, DIFFERENT SURFACES

GitHub Mobile and GitHub Desktop

GitHub Mobile

01 Triage and clear notifications

02 Read, comment on and edit issues and PRs

03 Edit files inside a pull request

04 Search users, repos and organisations

05 Supports 2FA and new-device verification

GitHub Desktop

01 Add, clone and publish repositories

02 Stage changes interactively and add co-authors

03 Check out PR branches and view CI status

04 Compare changed images visually

05 Open-source and free on Windows and macOS

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WRITE ONCE, RENDER EVERYWHERE ON GITHUB

GitHub-Flavored Markdown Essentials

Headings

through ###### set levels 1–6, mirroring HTML h1–h6.

Emphasis

italic or _italic_; **bold** or __bold__; escape a literal with a backslash.

Lists

1. for ordered, - or * for unordered; indent two spaces to nest.

Links and images

[text](url) links; the same syntax prefixed with ! embeds an image.

Tables and quotes

Pipes separate columns with a --- header row; > marks a block quote.

Code

Backticks for inline `code`; triple backticks plus a language name for a highlighted block.

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GFM EXTENDS STANDARD MARKDOWN

Markdown Features That Are GitHub-Specific

Task lists

- [x] and - [ ] render tickable checkboxes and a progress bar on issues.

Autolinked references

#123 links an issue or PR; owner/repo#123 crosses repositories.

Commit links

Paste seven or more characters of a commit SHA and GitHub links the commit.

@mentions

@user or @org/team notifies and records who was pulled into a thread.

Inline HTML

Use raw HTML such as
 or <details> where Markdown has no syntax.

Slash commands

/code, /table, /details, /tasklist, /saved-replies and /template insert Markdown for you.

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DOMAIN 1 · HANDS-ON

Lab 1: Install Git, Git Bash and GitHub Desktop

Northstar Learning is onboarding a new contributor on Windows or macOS and needs a repeatable setup record.

YOU WILL BUILD

A verified Git/Git Bash and GitHub Desktop installation with identity and default-branch configuration

https://git-scm.com/install/, https://desktop.github.com/download/, Git Bash or terminal, Excel evidence workbook

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 1 · SEE LG FOR PROCEDURES

Lab 1 Workflow and Acceptance

1. Install Git and open Git Bash
2. Configure identity and main
3. Install and sign in to Desktop
4. Verify tools and record evidence

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ Git version is recorded
- ✓ GitHub Desktop is signed in
- ✓ Git Bash or terminal works
- ✓ Identity and default branch are verified

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DOMAIN 1 · HANDS-ON

Lab 2: Create and Clone a Training Repository

The onboarding team needs a safe repository that every later lab can reuse.

YOU WILL BUILD

A public northstar-git-lab repository cloned locally with a verified origin remote

GitHub.com, Git Bash or terminal, GitHub Desktop, Markdown, Excel evidence workbook

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 2 · SEE LG FOR PROCEDURES

Lab 2 Workflow and Acceptance

1. Create the remote repository
2. Clone with git clone
3. Inspect origin and status
4. Edit and compare CLI with Desktop

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ Remote repository exists
- ✓ Local clone has origin
- ✓ CLI and Desktop show the same change
- ✓ Workbook records paths and states

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RETRIEVAL PRACTICE

Domain 01 Recap

Can you distinguish Git from GitHub and explain the GitHub Flow without relying on menu positions?

Exam readiness comes from recognising the purpose, trade-off and best-fit feature in a scenario.

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02

DOMAIN 02 · 10–15%

Work with GitHub Repositories

Repository structure · templates · files · branches · insights

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EXAM WEIGHTING 10–15%

Key Concepts — Work with GitHub Repositories

Repository health

README, LICENSE, CONTRIBUTING, CODEOWNERS and SECURITY establish trust.

Create and organise

Templates, branches and clear names make repositories repeatable.

Manage files

init, clone, status, diff, add, commit, log, branch, merge, fetch, pull and push.

Visibility

Insights, traffic, contributors, stars and dependency views reveal health.

Maintenance

Archive, transfer, protect and document repositories deliberately.

Feature previews

Test emerging capabilities without confusing preview with GA.

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REPOSITORY HEALTH

Five Files That Make a Repository Trustworthy

README

Purpose, use, setup and examples.

LICENSE

Permissions and conditions.

CONTRIBUTING

How to propose changes.

CODEOWNERS

Who reviews which paths.

SECURITY

How to report vulnerabilities.

.gitignore

What never belongs in commits.

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TWO STARTING PATHS

Start with init or clone?

git init

01 Existing local folder

02 Creates .git history store

03 No remote by default

04 Add files and first commit

git clone <url>

01 Existing remote repository

02 Copies files and history

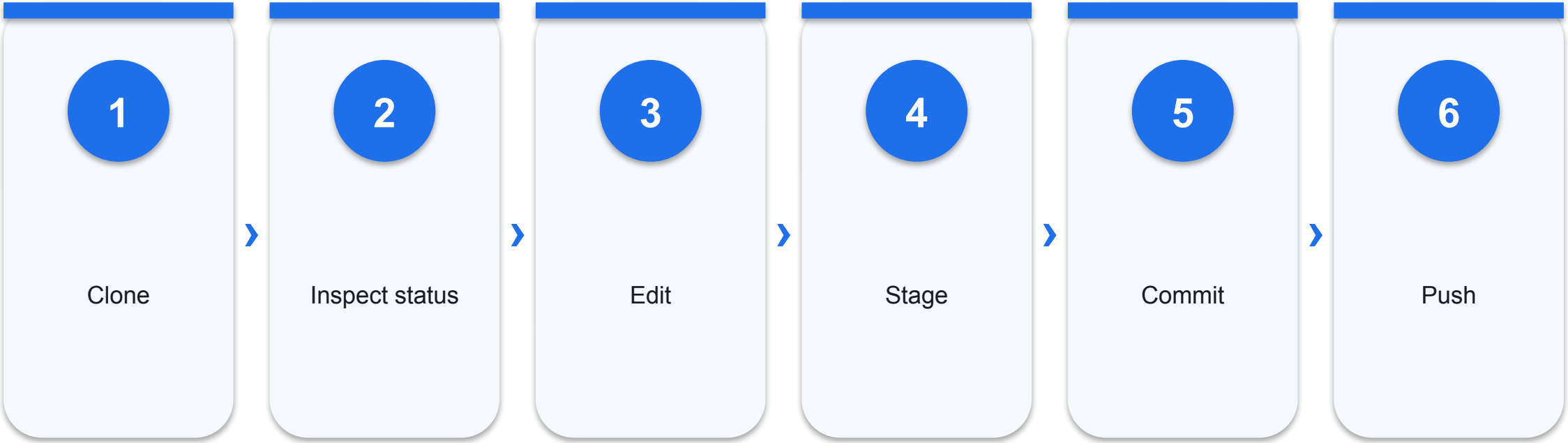
03 Configures origin

04 Checks out the default branch

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GIT WORKFLOW

Local Contribution Loop



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SMALL COMMITS TELL A CLEAR STORY

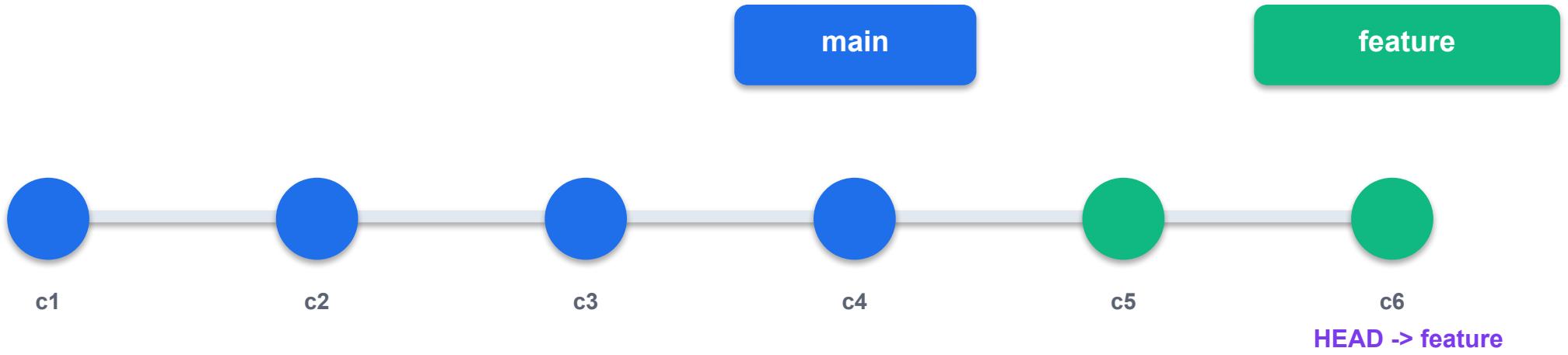
Selective Staging



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COMMIT GRAPH · HEAD · MAIN · FEATURE

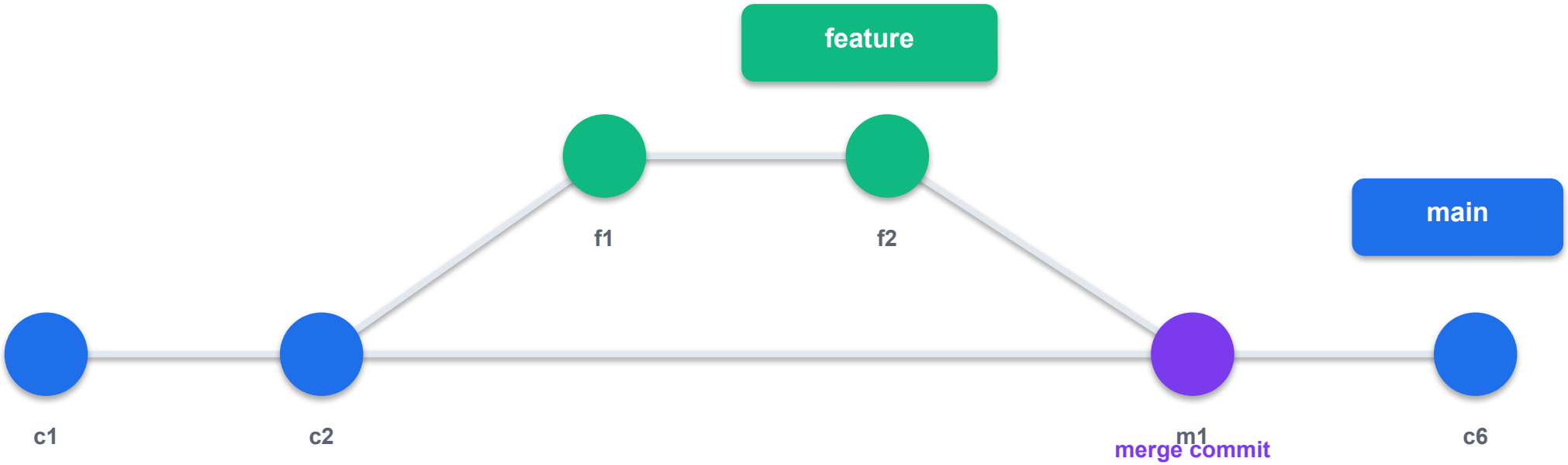
Branches Are Movable Pointers



A commit stays fixed. Committing advances the current branch pointer; switching changes which branch HEAD follows.

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Read a Real History Graph




```
git log --oneline --graph --all --decorate
```


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MERGE STRATEGIES

Fast-Forward or Merge Commit?

Fast-forward

01 Main has not diverged

02 Pointer advances

03 No new merge commit

04 Linear history

Three-way merge

01 Both lines have commits

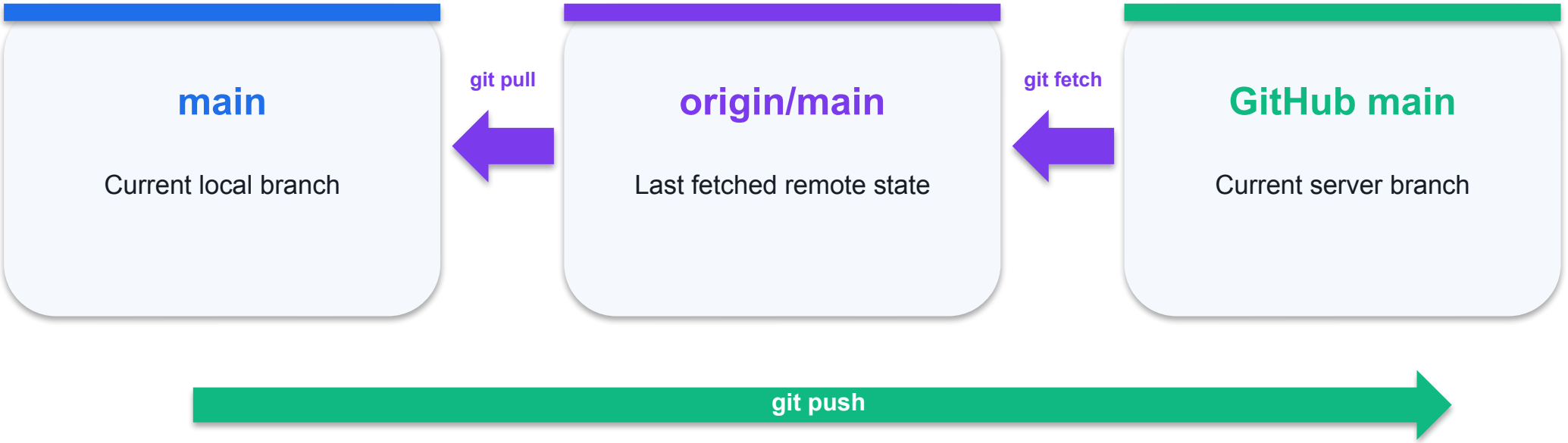
02 Two parents are combined

03 New merge commit

04 Preserves divergence context

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Local, Tracking and Remote Branches



Fetch updates origin/main only. Pull fetches and integrates. Push publishes local commits to the remote branch.

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SEPARATE OBSERVATION FROM INTEGRATION

Fetch Before Pull

git fetch origin

01 Downloads objects and refs

02 Updates origin/main

03 Leaves working files unchanged

04 Lets you inspect first

git pull --ff-only

01 Fetches remote work

02 Integrates current branch

03 Rejects unexpected divergence

04 Changes local branch/files

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-U CONNECTS LOCAL AND REMOTE BRANCHES

Publish a New Branch

1. Create branch
2. Commit locally
3. git push -u origin <branch>
4. Verify upstream
5. Open pull request

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CONFLICTS NEED A HUMAN DECISION

Resolve a Merge Conflict

1. Git stops
2. git status identifies paths
3. Choose final content
4. git add marks resolved
5. git commit completes merge
6. Push and verify

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The Three Trees and What Reset Moves

HEAD

The commit you are on

Last committed snapshot

Index (staging)

What the next commit will contain

Proposed snapshot

Working tree

The files on disk

Editable snapshot

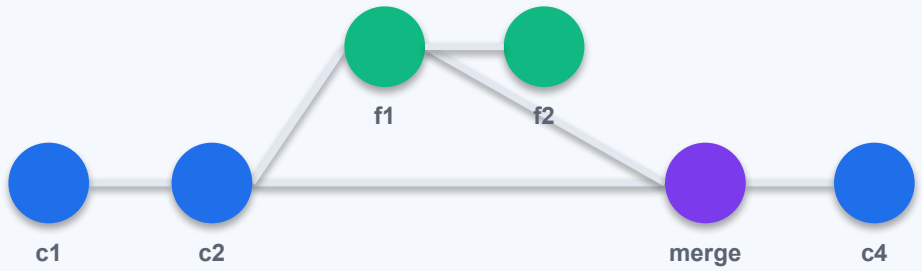
`git reset --soft <commit>`	**Moves HEAD only**	Index and working tree keep your work; the change is still staged.
`git reset --mixed <commit>`	**Moves HEAD + index**	Default. Work survives in the working tree but is unstaged.
`git reset --hard <commit>`	**Moves all three**	Destructive: uncommitted work is discarded. Recover via reflog.
`git restore <file>`	**Working tree only**	Discards one file's unstaged edit; history and index untouched.

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SAME WORK · TWO DIFFERENT GRAPHS

Merge Keeps History; Rebase Rewrites It

git merge feature



Both lines survive. The merge commit has two parents and records when and by whom the branches were joined.

git rebase main



Linear history. f1 and f2 are replayed as new commits with new SHAs, so the originals no longer exist.

Golden rule: never rebase commits that others have already pulled.

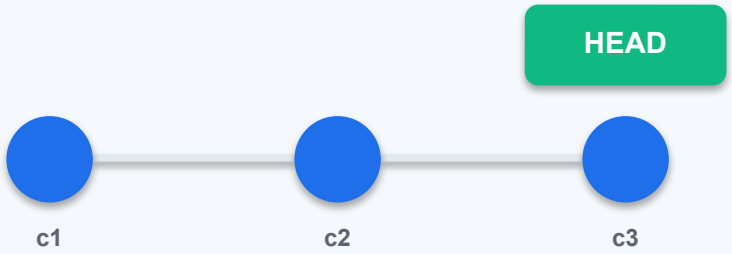
Rebase your own unpushed branch to tidy it before review; merge when the branch is shared. Squash-and-merge on GitHub gives a linear main without rewriting anyone's local history.

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HEAD POINTS AT A COMMIT, NOT A BRANCH

Detached HEAD: What It Is and How to Recover

Attached



HEAD → main → c3

Detached



HEAD → c2 directly

How you get there	git switch <sha>, git checkout <tag>, or opening a commit from a bisect or a CI checkout.
Why it matters	New commits belong to no branch. Switching away leaves them unreferenced and eventually collectable.
Keep the work	git switch -c rescue — names the current commit so it is safe.
Already switched away	git reflog finds the SHA, then git switch -c rescue <sha> restores it.

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THE DECIDING QUESTION IS SHARING, NOT SEVERITY

Reset or Revert? Decide by Who Has Seen It

git reset — local only

01 Rewrites which commit the branch points at

02 The old commits leave the branch

03 Fine on an unpushed local branch

04 Needs a force push if already shared — avoid

05 reflog is the safety net if you go too far

git revert — safe when shared

01 Adds a new commit that undoes an earlier one

02 Nothing is rewritten; both commits remain

03 Colleagues pull it like any other commit

04 The audit trail shows what was undone and why

05 The correct choice for anything already on main

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A CLEAN TREE WITHOUT AN UNFINISHED COMMIT

Park Work in Progress with git stash

Why stash

An urgent fix arrives while your tree is dirty and you need a clean branch now.

Park it

git stash push -m "wip" saves tracked modifications and restores a clean tree.

Include untracked

git stash push -u also parks new files that Git is not yet tracking.

See what you have

git stash list shows entries newest-first as stash@{0}, stash@{1} and so on.

Bring it back

git stash pop reapplies the newest entry and drops it; apply keeps it stored.

Caution

A stash is local and never pushed. It is a short-lived parking space, not a backup.

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Annotated or Lightweight Tag?

Annotated — use for releases

01 git tag -a v1.0.0 -m "Release 1.0.0"

02 A real Git object with tagger, date and message

03 Can be signed and verified

04 What GitHub Releases are built from

05 Push explicitly: git push origin --tags

Lightweight — a private bookmark

01 git tag v1.0.0-light

02 Just a name pointing at a commit

03 No author, date or message is stored

04 Fine for a temporary local marker

05 Not appropriate for a published release

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SURGICAL CHANGES TO HISTORY

Move or Repair Single Commits

git cherry-pick <sha>

Copies one commit's change onto the current branch as a new commit with a new SHA.

When to cherry-pick

Backport one fix to a release branch when merging the whole branch would bring unwanted work.

git commit --amend

Replaces the newest commit — message, content or both — producing a different SHA.

Amend safely

Only before pushing. Amending shared history forces everyone else to reconcile a rewritten commit.

Duplicate risk

Cherry-picking and then merging the same branch can present the change twice; prefer one route.

Verify after

git log --oneline and git show confirm what actually landed before you push.

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ALMOST NOTHING IS TRULY LOST

git reflog Is the Safety Net

What it records

Every position HEAD has held locally — commits, switches, merges, resets and rebases.

Why it saves you

A reset --hard or a lost detached commit is still listed with its SHA for about 90 days.

Read it

git reflog prints entries as HEAD@{0}, HEAD@{1} with the action that moved HEAD.

Recover

git switch -c rescue <sha> names the lost commit so it can never be collected.

Local only

The reflog is never pushed or cloned — it protects your machine, not the team.

Discipline

Note the current SHA before a destructive command; verify with git log afterwards.

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IGNORE EARLY; UNTRACK EXPLICITLY

.gitignore Mechanics and Its One Trap

Pattern rules

01 One pattern per line; blank lines and # comments are ignored

02 build/ ignores a directory; *.log ignores an extension

03 A leading / anchors the pattern to the repository root

04 A leading ! re-includes a previously excluded path

05 Language templates live at github.com/github/gitignore

The trap: already-tracked files

01 gitignore only affects files Git is not yet tracking

02 A committed secret or build artefact keeps being committed

03 git rm --cached <file> untracks it while keeping it on disk

04 Commit that removal and the ignore rule finally applies

05 A leaked secret is still in history — revoke and rotate it

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FIND THE COMMIT THAT EXPLAINS THE LINE

Investigate History: log Filters and blame

git log --since="2 weeks"

Limit history to a time window; combine with --until for a closed range.

git log --author=<name>

Show only one contributor's commits when tracing a change.

git log -- <path>

Show only the commits that touched one file or directory.

git log -S"<text>"

Pickaxe search: find the commit that introduced or removed a string.

git blame <file>

Annotate every line with the commit, author and date that last changed it.

Blame on GitHub

The web Blame view adds per-line history navigation and links to each pull request.

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DOMAIN 2 · HANDS-ON

Lab 3: Practice Status, Diff, Add, Commit and Log

The repository needs small, reviewable commits rather than one uncontrolled upload.

YOU WILL BUILD

A three-commit repository history with selective staging, clear messages and command evidence

Git Bash or terminal, GitHub Desktop, text editor, GitHub.com, Excel command workbook

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 3 · SEE LG FOR PROCEDURES

Lab 3 Workflow and Acceptance

1. Inspect status and diff
2. Stage files or selected hunks
3. Commit focused snapshots
4. Push and verify history

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ Three focused commits exist
- ✓ Staged content was reviewed
- ✓ Amend and stash effects are evidenced
- ✓ History matches locally and remotely
- ✓ Command log records actual outputs

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DOMAIN 2 · HANDS-ON

Lab 4: Create Branches and Merge Changes

A navigation improvement must be developed without destabilising the main branch.

YOU WILL BUILD

A feature branch, separate commit history, local merge, annotated release tag and readable commit graph

Git Bash or terminal, GitHub Desktop branch visualisation, GitHub Releases, Excel evidence workbook

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 4 · SEE LG FOR PROCEDURES

Lab 4 Workflow and Acceptance

1. Create and switch feature branch
2. Make two feature commits
3. Switch to main and merge
4. Tag the release and inspect the graph

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ Feature work is isolated
- ✓ Merge result is explained
- ✓ Rebase safety rule is recorded
- ✓ Annotated tag exists locally and on GitHub
- ✓ Merged branch is safely deleted

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DOMAIN 2 · HANDS-ON

Lab 5: Synchronise with Fetch, Pull and Push

A teammate changed the remote README while the learner was working locally.

YOU WILL BUILD

A synchronised local and remote repository with tracking-branch evidence

Git Bash or terminal, GitHub.com, GitHub Desktop Fetch/Pull/Push controls, Excel workbook

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 5 · SEE LG FOR PROCEDURES

Lab 5 Workflow and Acceptance

1. Simulate a remote commit
2. Fetch and inspect origin/main
3. Pull with fast-forward only
4. Push a branch and verify tracking

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ Fetch is distinguished from pull
- ✓ Incoming commit is inspected
- ✓ Upstream tracking is verified
- ✓ Local and remote histories match

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DOMAIN 2 · HANDS-ON

Lab 6: Resolve a Merge Conflict and Recover Safely

Two contributors changed the same README sentence and Git cannot decide which version to keep.

YOU WILL BUILD

A resolved merge conflict plus a tested restore, unstage, reset, revert and reflog recovery decision record

Git Bash or terminal, GitHub.com, GitHub Desktop conflict helper, text editor, Excel workbook

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 6 · SEE LG FOR PROCEDURES

Lab 6 Workflow and Acceptance

1. Create overlapping local and remote edits
2. Pull to trigger the conflict
3. Resolve, add and commit
4. Practise reset, revert and reflog recovery

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ Conflict markers are removed
- ✓ Merge completes without lost history
- ✓ Reset variants are distinguished by effect
- ✓ A discarded commit is recovered via reflog
- ✓ Resolution and revert are evidenced

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RETRIEVAL PRACTICE

Domain 02 Recap

Can you explain what makes a GitHub repository healthy, maintainable and trustworthy?

Exam readiness comes from recognising the purpose, trade-off and best-fit feature in a scenario.

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03

DOMAIN 03 · 10–15%

Collaborate Using GitHub

Issues · pull requests · discussions · notifications · Pages

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EXAM WEIGHTING 10–15%

Key Concepts — Collaborate Using GitHub

Choose the right object

Issue for tracked work, discussion for conversation, PR for proposed change.

Traceability

Closing keywords link PRs to issues and preserve decision context.

Review

Comments, suggestions, approvals and required reviewers protect quality.

Triage

Labels, assignments, filters, templates and notifications keep work actionable.

Share knowledge

Gists, Wikis and Pages publish snippets, documentation and sites.

Fork workflow

Fork, branch, change and upstream PR enable external contribution.

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Issue, Discussion or Pull Request?

Conversation and work

01 Issue: track actionable work

02 Discussion: ask, share and decide

03 Notification: route attention

04 Milestone: group an outcome

Proposed change

01 PR compares branches

02 Review comments preserve context

03 Checks provide evidence

04 Merge records the decision

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END-TO-END TRACEABILITY

Traceable Change

1. Issue defines outcome
2. Branch isolates work
3. PR uses closing keyword
4. Review improves change
5. Merge closes issue

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Commands to Pull Request

1. git switch -c <branch>
2. git add + git commit
3. git push -u origin <branch>
4. Open linked PR
5. Review + required checks
6. Merge PR, then pull main

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KNOWLEDGE SHARING

Share Beyond the Repository

Gist

Share a small versioned snippet.

Wiki

Maintain deeper repository documentation.

Pages

Publish a static site from a repository.

Notifications

Subscribe at the right level.

Templates

Capture consistent issue or PR context.

Fork

Contribute without direct write access.

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ATTENTION IS THE SCARCE RESOURCE

Notifications and Subscriptions

What you can subscribe to

01 A single conversation in an issue, PR or gist

02 CI activity such as Actions workflow status

03 Repository events: issues, PRs, releases, security alerts, discussions

04 All activity in a repository

05 A team you belong to, via @org/team mentions

Choosing the right level

01 All activity: only for repositories you maintain

02 Participating and @mentions: the sustainable default

03 Custom: pick releases or security alerts only

04 Ignore: silence a noisy repository entirely

05 Review the inbox filters before adding more subscriptions

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AUTOLINKED REFERENCES

Cross-Linking Builds Context Automatically

Issue references

#123 in any comment links that issue or pull request in the same repository.

Across repositories

owner/repo#123 links an item in another repository, including a branch reference.

Commit references

Pasting seven or more characters of a commit SHA autolinks to that commit.

Closing keywords

Closes, Fixes or Resolves #123 in a PR body closes the issue when the PR merges.

Automatic links

Opening a PR from a branch links its commits without any manual action.

Manual links

The PR sidebar links issues and projects that the text did not mention.

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TWO SEARCH BOXES, TWO PURPOSES

Find Things: Global vs Context Search

Global search

01 Search box at the top of every page

02 Spans all of GitHub — code, issues, users, Marketplace

03 Full search syntax with qualifiers

04 Best for discovery across many repositories

05 Example: language:python topic:devops stars:>100

Context search

01 The search box inside a repository tab

02 Scoped to that repository and that item type

03 Suggests authors, labels, milestones and projects

04 Best for triage inside a project you know

05 Example: is:issue is:open label:bug assignee:@me

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TRIAGE IS A REPEATABLE PROCESS

Respond Consistently at Scale

Saved replies

Store a standard response once and insert it with /saved-replies or the reply menu.

Issue templates

Give reporters a structured form so triage does not start with a question.

PR templates

Prompt every author for verification evidence and linked issues.

Labels

Encode type, priority and status so filters and automation can act on them.

@mentions

Pull a specific person or @org/team into a thread and leave a record of who was asked.

Code review

Comments, suggested changes and required reviewers turn feedback into commits.

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DOMAIN 3 · HANDS-ON

Lab 7: Open a Pull Request and Complete Code Review

A learner onboarding checklist requires peer review and issue traceability before release.

YOU WILL BUILD

A linked issue, pushed branch, reviewed pull request, controlled merge and local clean-up

Git Bash, GitHub pull requests, GitHub Desktop, Excel evidence workbook

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 7 · SEE LG FOR PROCEDURES

Lab 7 Workflow and Acceptance

1. Create issue and feature branch
2. Commit and push with upstream
3. Open linked PR and complete review
4. Merge, pull main and clean up

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ PR links to issue
- ✓ Main is refreshed locally
- ✓ Review changes the branch
- ✓ Both branch copies are cleaned up

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DOMAIN 3 · HANDS-ON

Lab 8: Coordinate Work with Issues, Discussions and Pages

Northstar needs structured bug intake, a Q&A area and a lightweight project site.

YOU WILL BUILD

Issue form, triaged and cross-linked issue, Q&A discussion, notification plan and GitHub Pages site

Issues, Discussions, GitHub-Flavored Markdown, notifications, Gists, Wikis, GitHub Pages, Excel workbook

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 8 · SEE LG FOR PROCEDURES

Lab 8 Workflow and Acceptance

1. Create labels and templates
2. Open issue and discussion with rich Markdown
3. Cross-link, mention and enable Pages
4. Tune notifications and record evidence

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ Issue is triaged
- ✓ Markdown features render as intended
- ✓ Cross-links and @mentions resolve
- ✓ Notification choice is justified
- ✓ Pages URL resolves

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RETRIEVAL PRACTICE

Domain 03 Recap

Can you choose among issues, discussions and pull requests for a collaboration scenario?

Exam readiness comes from recognising the purpose, trade-off and best-fit feature in a scenario.

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04

DOMAIN 04 · 10–15%

Apply Modern Development Practices

Actions · Copilot · Codespaces · dev containers · github.dev

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EXAM WEIGHTING 10–15%

Key Concepts — Apply Modern Development Practices

GitHub Actions

Events trigger workflows made of jobs, steps, actions and runners.

Continuous feedback

Automated checks shorten the gap between change and evidence.

GitHub Copilot

AI-assisted suggestions, chat, agents and multi-model support accelerate work.

Plan differences

Individual, Business and Enterprise plans add progressively stronger governance.

Codespaces

Cloud development environments standardise compute and dependencies.

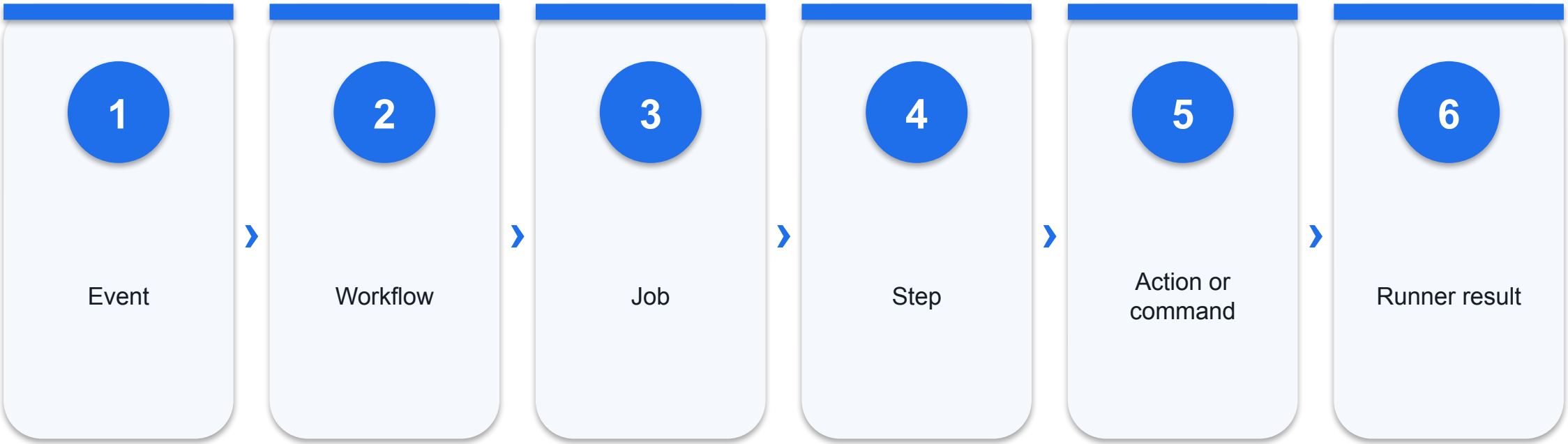
github.dev

A lightweight browser editor; Codespaces provides a full runtime and terminal.

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AUTOMATION MODEL

An Actions Workflow



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CHOOSE THE RIGHT ENVIRONMENT

github.dev vs Codespaces

github.dev

01 Instant browser editor

02 No terminal or runtime

03 Good for small edits

04 Uses repository files

Codespaces

01 Cloud VM and container

02 Terminal and runtime

03 Reproducible dev container

04 Good for build and test

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EVENT → WORKFLOW → JOB → STEP

Anatomy of a Workflow File

Location

A YAML file in .github/workflows/ on the branch that should run it.

name and on

name labels the workflow; on lists the events that trigger it.

jobs

Named units of work; each job selects a runner with runs-on.

steps

Ordered actions or shell commands, sharing the job's workspace.

uses vs run

uses calls a published action such as actions/checkout@v4; run executes a shell command.

Reading a failure

Open the run, then the job, then the failing step; the log names the exact command and exit code.

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AUTOMATION HAS A COST MODEL

Triggers and Actions Billing

Common triggers

01 push — validate every commit on a branch

02 pull_request — gate a change before merge

03 schedule — cron for nightly or weekly runs

04 workflow_dispatch — a manual Run workflow button

05 issues, release, and other repository events

Billing basics

01 Free for public repositories and self-hosted runners

02 Private repositories get a monthly free minute allowance by plan

03 Minutes reset monthly; storage accumulates

04 Spending limit defaults to $0 for card-billed accounts

05 Larger runners consume minutes at a higher multiplier

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AI PAIR PROGRAMMER, NOT AN AUTOPILOT

What GitHub Copilot Actually Offers

Inline suggestions

Greyed-out completions ahead of the cursor; Tab accepts, Esc dismisses.

Copilot Chat

A conversational panel that explains code, proposes fixes and drafts tests.

Inline chat

Ctrl+I or Cmd+I asks for a change at the exact cursor position.

Copilot in the CLI

Suggests and explains shell commands in Bash, Zsh and PowerShell.

Pull request summaries

Drafts a PR description from the actual diff for the author to edit.

Where it runs

VS Code, Visual Studio, JetBrains IDEs, Neovim, github.com and Mobile.

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CAPABILITY AND GOVERNANCE SCALE TOGETHER

Copilot Plan Tiers and Interaction

Plan tiers

01 Free: a capped monthly allowance of completions and chat

02 Pro: individual developers, full completions and chat

03 Business: policy control, public-code filtering, IP indemnity

04 Enterprise: codebase indexing, docs search, PR review assistance

05 Organisation owners enable and govern seats centrally

How you interact

01 Comment to code: describe the intent and let it draft

02 Multiple suggestions: cycle alternatives and pick one

03 Explain this: understand unfamiliar code before changing it

04 Generate unit tests for a selected function or class

05 Every suggestion still needs human review before commit

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PROMPT QUALITY DETERMINES OUTPUT QUALITY

Prompting and Troubleshooting Copilot

Start simple

Ask for the basic shape first, then refine with a follow-up instruction.

Be specific

Name the framework, the input and the expected output — vague prompts return vague code.

Give context

Open the related files; Copilot reads the surrounding code and comments.

Iterate in chat

Use chat to correct an almost-right suggestion rather than retyping the prompt.

Enable or disable

The status bar icon toggles completions globally or for one language.

Troubleshoot

Collect Diagnostics from the command palette; check logs, proxies and firewall rules.

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CREATE → USE → STOP → REBUILD → DELETE

The Codespace Lifecycle

1. Create from repo, branch, PR or commit
2. Work — autosave in the browser
3. Stop or time out after 30 min idle
4. Reopen and continue
5. Rebuild to apply dev-container changes
6. Delete after pushing

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PUSH BEFORE YOU DELETE

Codespaces: Persistence and Personalisation

Creation process

A VM and storage are assigned, a container is built, a connection opens, then post-create setup runs.

Saving work

Autosave protects the VM only. Committing and pushing is the only durable save.

Timeout and cost

30 minutes idle stops it by default; running time bills CPU, stopped time bills storage.

Rebuild behaviour

Changes inside /workspaces survive a rebuild; changes outside it are cleared.

Deletion

Stopped codespaces are auto-deleted after a retention period, 30 days by default.

Personalisation

Dotfiles repository, Settings Sync, extensions, shell, machine type, region and timeout.

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DOMAIN 4 · HANDS-ON

Lab 9: Build and Diagnose a GitHub Actions Check

Every documentation pull request must prove that README.md and SECURITY.md exist.

YOU WILL BUILD

A pull-request Actions workflow with successful and failed run evidence

GitHub Actions, YAML, Git Bash, workflow logs, Excel workbook

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 9 · SEE LG FOR PROCEDURES

Lab 9 Workflow and Acceptance

1. Create the validation workflow
2. Push branch and open PR
3. Diagnose a failed run and fix
4. Rerun and capture passing logs

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ Workflow triggers on PR
- ✓ Corrected run passes
- ✓ Failure is diagnosed from logs
- ✓ Run URLs and SHAs are captured

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DOMAIN 4 · HANDS-ON

Lab 10: Compare Codespaces, github.dev and Copilot

A distributed contributor needs a consistent environment for a small documentation automation task.

YOU WILL BUILD

A dev-container configuration, environment comparison and verified Copilot-assisted change

Codespaces, dev containers, github.dev, GitHub Copilot or trainer demonstration, Excel workbook

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 10 · SEE LG FOR PROCEDURES

Lab 10 Workflow and Acceptance

1. Edit in github.dev
2. Open Codespace and verify runtime
3. Use Copilot with human review
4. Commit verified change and compare tools

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ Dev container is valid
- ✓ AI output is independently checked
- ✓ Environment choice is justified
- ✓ Verified change is committed

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RETRIEVAL PRACTICE

Domain 04 Recap

Can you select the right GitHub automation, AI and cloud-development service for a scenario?

Exam readiness comes from recognising the purpose, trade-off and best-fit feature in a scenario.

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05

DOMAIN 05 · 5–10%

Manage Projects with GitHub

Projects · views · fields · workflows · milestones · insights

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EXAM WEIGHTING 5–10%

Key Concepts — Manage Projects with GitHub

One data model

Issues, PRs and draft items can be viewed as table, board or roadmap.

Fields

Status, iteration, priority and custom fields turn work into structured data.

Automation

Built-in workflows update status as items open, close or merge.

Planning

Milestones and iterations express target outcomes and timeboxes.

Communication

Assignees and saved replies make ownership and responses consistent.

Insights

Charts expose flow, workload, progress and bottlenecks.

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PROJECTS

One Project, Three Views

Table

Compare many fields and edit in bulk.

Board

Visualise flow by status or another field.

Roadmap

Plan work against dates and iterations.

Fields

Create shared structure.

Workflows

Automate predictable transitions.

Insights

Turn project data into decisions.

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SCOPE · TIME · OWNERSHIP

Release Coordination

1. Define milestone
2. Prioritise items
3. Assign owners
4. Track required checks
5. Tag and publish release

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ONE DATASET, MANY VIEWS

Projects vs Projects Classic

Projects (current)

01 Table, board and roadmap layouts over one dataset

02 Custom fields: text, number, date, iteration, single select

03 Tokenised filtering and saved views

04 Insights charts with aggregation functions

05 GraphQL ProjectsV2 API and webhook events

Projects Classic (legacy)

01 Board columns only, per repository or organisation

02 Note cards with no structured fields

03 No iteration planning or roadmap view

04 Limited automation presets

05 Retained for existing boards; new work uses Projects

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STRUCTURE BEFORE AUTOMATION

Fields Turn Work into Data

Status

The single-select field that drives board columns and built-in workflows.

Priority

A single-select field such as Urgent, High, Medium, Low for grouping and slicing.

Iteration

Repeating timeboxes with a start date, duration and optional breaks for sprints or quarters.

Date and number

Target date and estimate fields feed the roadmap and insight aggregations.

Grouping

Group a table or board by any field to answer a different stakeholder question.

Slicing

The slice panel filters the whole view by one field's values without editing the filter.

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SEE THE FLOW, THEN AUTOMATE IT

Insights and Built-in Workflows

Insights

01 Current charts show the project as it is right now

02 Historical charts show trends over time (Team and Enterprise Cloud)

03 Burn-up is the default historical chart

04 Bar, column, line and stacked area layouts

05 Sum, count, average, min and max aggregations

Built-in workflows

01 Item added to project → set Status to Todo

02 Item closed → set Status to Done

03 Pull request merged → set Status to Done

04 Auto-archive items matching a filter

05 Beyond these: the GraphQL API or an Actions workflow

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WHO CAN SEE IT AND WHAT HAPPENS AT THE END

Project Access and Lifecycle

Organisation projects

Base access of No access, Read, Write or Admin for everyone in the organisation.

User projects

Access is granted per invited collaborator as Read, Write or Admin.

Visibility

Private or public, changed in Settings under the Danger zone.

Link to a team

Team → Projects → Link a project makes it visible in that team's workspace.

Link to a repository

Repository → Projects → Link a project surfaces it where the work lives.

Close vs delete

Closing archives the project but keeps its data; deleting is permanent and requires typing the name.

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DOMAIN 5 · HANDS-ON

Lab 11: Plan and Track a Release with GitHub Projects

The CourseTrack Pro team needs one view of features, defects and release readiness.

YOU WILL BUILD

A GitHub Project with table, board and roadmap views plus fields, workflows and insights

GitHub Projects, issues, milestones, labels, saved replies, Excel workbook

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 11 · SEE LG FOR PROCEDURES

Lab 11 Workflow and Acceptance

1. Create Project and fields
2. Add issues and milestone
3. Build table, board and roadmap
4. Automate status and inspect insights

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ Three views answer different questions
- ✓ Workflow is enabled
- ✓ Fields are populated
- ✓ Insight is interpreted

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RETRIEVAL PRACTICE

Domain 05 Recap

Can you choose the right GitHub Projects view, field and workflow for a planning question?

Exam readiness comes from recognising the purpose, trade-off and best-fit feature in a scenario.

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06

DOMAIN 06 · 10–15%

Understand Privacy, Security and Administration

2FA · passkeys · roles · visibility · rulesets · EMU · policy

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EXAM WEIGHTING 10–15%

Key Concepts — Understand Privacy, Security and Administration

Secure identity

2FA and passkeys reduce account-takeover risk.

Least privilege

Repository and organisation roles should match responsibility.

Visibility

Public, private and internal determine who can discover and access work.

Protected change

Rulesets, branch protection, required reviews and checks guard critical branches.

Enterprise control

EMUs and organisation policies centralise identity and Copilot governance.

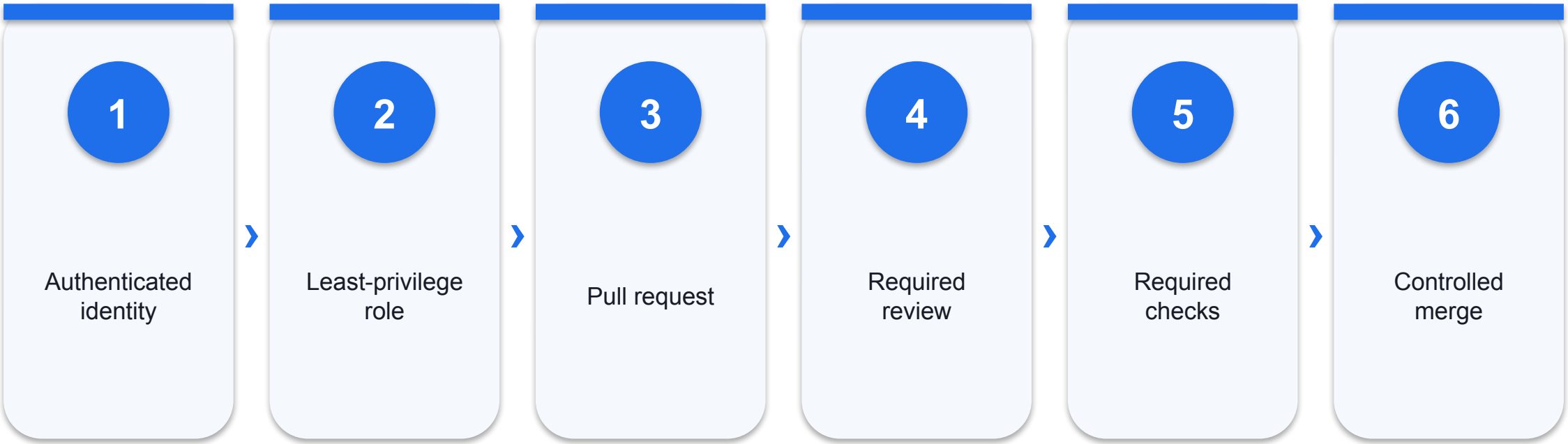
Secure supply chain

Dependabot, code scanning and secret scanning identify different risks.

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SECURE COLLABORATION

Protected Change Path



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PICK THE NARROWEST CREDENTIAL THAT WORKS

The Authentication Landscape

Username + password

No longer accepted for Git operations over HTTPS; sign-in only, and always with 2FA.

Classic PAT

One token, broad scopes, account-wide reach. Simple but blunt — set an expiry.

Fine-grained PAT

Scoped to chosen repositories with per-permission control and a required expiry. Prefer this.

SSH key

A key pair on your machine; no token to paste per push. Protect it with a passphrase.

Deploy key

An SSH key granting one repository, read-only by default — the right choice for a CI server.

GitHub App / OAuth

Delegated, revocable access for integrations rather than a human's credentials.

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Second Factors and Enterprise Identity

2FA options, strongest first

01 Passkeys and security keys: phishing-resistant

02 TOTP app: reliable, backs up to the cloud

03 SMS: last resort, not available everywhere

04 Recovery codes: store offline, never in a repo

05 Owners can require 2FA org-wide or enterprise-wide

Enterprise identity

01 SAML SSO: the IdP authenticates, GitHub authorises

02 SCIM: automatic provisioning and deprovisioning

03 LDAP: directory-backed accounts on Enterprise Server

04 Team synchronisation: IdP groups drive team membership

05 EMU: the enterprise owns the accounts entirely

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LEAST PRIVILEGE IS A CHOICE PER PERSONA

Permission Levels You Must Recognise

Repository

Read → Triage → Write → Maintain → Admin; each level includes the ones below it.

Triage in practice

Manage issues and pull requests without any ability to push code.

Maintain in practice

Run the repository day to day without destructive or security-sensitive settings.

Team

Member or maintainer; teams then carry a repository permission to their members.

Organisation

Owner, member, moderator and billing manager, plus security manager.

Enterprise

Enterprise owner, member and billing manager, setting policy across all organisations.

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FIND VULNERABILITIES BEFORE MERGE

Code Scanning with CodeQL

Setting it up

01 Default setup: GitHub picks languages and queries

02 Advanced setup: a CodeQL workflow file you control

03 Third-party tools upload SARIF 2.1.0 results

04 Free for public repositories; Advanced Security for private

05 Alerts land in the repository Security tab

Tuning what runs

01 on: push validates protected branches

02 on: pull_request scans the merge commit before merge

03 on: schedule keeps scanning idle repositories

04 paths and paths-ignore skip irrelevant changes

05 A severity threshold decides which alerts fail the PR check

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THREE DEPENDABOT FEATURES, THREE JOBS

Supply Chain and Secret Protection

Dependency graph

Parses manifests such as package.json to map direct and transitive dependencies.

Dependabot alerts

Cross-references that graph against GitHub Security Advisories and notifies you.

Dependabot security updates

Opens a pull request that bumps a vulnerable dependency to a safe version.

Dependabot version updates

Configured in .github/dependabot.yml to keep dependencies current on a schedule.

Secret scanning

Detects committed credentials and notifies the issuing provider so they can revoke.

Push protection

Blocks the push that would introduce a detected secret, before it ever reaches history.

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CONTROLS THAT SURVIVE STAFF CHANGES

Govern the Repository Itself

Policy files

01 SECURITY.md: how to report a vulnerability privately

02 Security advisories: fix in private, then publish to the CVE list

03 CODEOWNERS: required reviewers per path

04 CONTRIBUTING and CODE_OF_CONDUCT: expected behaviour

05 .gitignore: what must never be committed

Protected change

01 Rulesets and branch protection guard the default branch

02 Require a pull request, approvals and passing checks

03 Require signed commits or a linear history

04 Restrict who may push; lock a branch read-only

05 Removing a leaked secret means rewriting history and rotating it

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Personal Accounts and EMUs

Personal account model

01 User controls identity

02 Can work across organisations

03 Organisation applies access policies

04 Offboarding removes org access

Enterprise Managed Users

01 Enterprise provisions identity

02 Activity constrained to enterprise

03 Central lifecycle and policy

04 Designed for controlled environments

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DOMAIN 6 · HANDS-ON

Lab 12: Secure and Administer a Repository

The release repository must prevent unsafe changes and surface security findings early.

YOU WILL BUILD

A repository security baseline with authentication comparison, access rationale, ruleset design, CODEOWNERS and scanning evidence

Authentication settings, PAT and SSH key configuration, security settings, Dependabot, code scanning, secret scanning concepts, roles, rulesets, Excel workbook

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 12 · SEE LG FOR PROCEDURES

Lab 12 Workflow and Acceptance

1. Compare authentication methods
2. Review roles and visibility
3. Configure ruleset, CODEOWNERS and checks
4. Compare security controls and evidence

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ Authentication methods are compared with a chosen default
- ✓ Least privilege is justified
- ✓ Branch rule and CODEOWNERS are documented
- ✓ Security features are distinguished
- ✓ No real secret, token or key is recorded

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RETRIEVAL PRACTICE

Domain 06 Recap

Can you match GitHub identity, access and security controls to the risks they reduce?

Exam readiness comes from recognising the purpose, trade-off and best-fit feature in a scenario.

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07

DOMAIN 07 · 5–10%

Explore the GitHub Community

Open source · Sponsors · Marketplace · InnerSource · discovery

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EXAM WEIGHTING 5–10%

Key Concepts — Explore the GitHub Community

Open source

Shared licences and transparent collaboration compound public value.

Healthy contribution

CONTRIBUTING, codes of conduct and issue templates set expectations.

Sustainability

GitHub Sponsors helps fund maintainers and projects.

Discovery

Follow users and organisations; use topics, stars and search to find work.

Marketplace

Apps and Actions extend GitHub workflows.

InnerSource

Open-source practices improve reuse and collaboration inside an organisation.

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COMMUNITY PRACTICES

Open Source and InnerSource

Open source

01 Public discovery

02 Licence defines use

03 External community

04 Sponsors can fund maintainers

InnerSource

01 Organisation-only discovery

02 Internal reuse

03 Trusted contributor model

04 Open practices behind access control

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Healthy Community Signals

Licence

Clear permission to use and contribute.

README

Fast orientation.

Contributing guide

Expected workflow.

Code of conduct

Behavioural standard.

Topics and stars

Discovery signals.

Sponsors

Sustainability mechanism.

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DISCOVERY IS A SKILL

Find a Project You Can Actually Help

Start where you already are

The easiest first contribution is to a tool you already use and understand.

github.com/explore

Curated topics, trending repositories and collections tuned to your interests.

Search with qualifiers

topic:, language:, stars:> and good-first-issues:>1 narrow the field quickly.

The /contribute page

Every repository has one; it surfaces beginner-friendly issues automatically.

Entry-point labels

good first issue, help wanted and documentation mark low-context work.

Small first

Broken links, typos and outdated docs are real contributions and build trust.

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AVOID DUPLICATED AND REJECTED WORK

Communicate Intent Before You Code

1. Read README, CONTRIBUTING and the code of conduct
2. Check assignees and linked PRs on the issue
3. Comment to claim it, or open a new issue
4. Get maintainer agreement for anything large
5. Fork, branch and implement
6. Open the PR and respond to review

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CONTRIBUTION IS MORE THAN CODE

Sustain the Projects You Depend On

GitHub Sponsors

Recurring or one-off funding for maintainers and organisations.

How to sponsor

The Sponsor button on an eligible project; choose a tier and whether it is public.

Non-financial support

Triage, documentation, mentoring, design and translation all reduce maintainer load.

Why it matters

Most maintenance is volunteered; funding and help decide whether a project survives.

Marketplace

Apps and Actions extend GitHub, many published by the same community.

Give credit

Cite dependencies in your README and report bugs upstream rather than only patching locally.

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Run an InnerSource Programme

Make repositories usable

01 Internal visibility so any employee can discover it

02 Read, triage, write, maintain, admin applied per team

03 Descriptive name and a one-line description

04 A README that states purpose, setup and demo

05 A licence, even internally, to settle reuse questions

Make contribution predictable

01 CONTRIBUTING.md defines the process

02 Issue and PR templates capture the right detail up front

03 A stated branching, release and review workflow

04 Measure process: review turnaround, PR size, time to open

05 Measure teams: unique contributors, reuse, cross-team @mentions

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DOMAIN 7 · HANDS-ON

Lab 13: Contribute through a Fork and Explore the Community

A learner will propose a small documentation improvement to a training repository.

YOU WILL BUILD

A fork, upstream remote, contribution branch and community-health evaluation

Forks, Git Bash, pull requests, licences, Sponsors, Marketplace, topics, Excel workbook

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 13 · SEE LG FOR PROCEDURES

Lab 13 Workflow and Acceptance

1. Fork and clone the fork
2. Add upstream and fetch
3. Branch, commit and push
4. Open upstream PR and review community health

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ Origin/upstream are distinguished
- ✓ Branch is published to the fork
- ✓ Contribution follows project guidance
- ✓ Community signals are evaluated

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DOMAIN 7 · HANDS-ON

Lab 14: Build a GH-900 Readiness Plan

The learner has completed the practical work and needs a targeted final revision plan.

YOU WILL BUILD

A weighted GH-900 readiness dashboard and two-day revision plan

Official Microsoft practice assessment, Tertiary practice exam, Excel readiness dashboard

EVIDENCE

Excel workbook

Action log · URLs · screenshots · status · reflection

Detailed procedures are in the Learner Guide and lab folder.

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LAB 14 · SEE LG FOR PROCEDURES

Lab 14 Workflow and Acceptance

1. Take a timed practice assessment
2. Enter domain scores
3. Prioritise weighted gaps
4. Plan revision and schedule the retest

ACCEPTANCE CRITERIA — THE LAB IS COMPLETE ONLY WHEN EVERY ITEM IS EVIDENCED

- ✓ Domain scores are entered
- ✓ Study plan links to hands-on evidence
- ✓ Priorities use weighted gaps
- ✓ Exam-day requirements are confirmed

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RETRIEVAL PRACTICE

Domain 07 Recap

Can you explain how to discover, contribute to and sustain a healthy GitHub community?

Exam readiness comes from recognising the purpose, trade-off and best-fit feature in a scenario.

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GIT

APPENDIX

Git Command Cheat Sheet

Every command used in this course, grouped by the job it does

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Cheat Sheet 1 — Setup and Snapshot

Install, identify yourself, start a repository and record snapshots.

git --version
Verify that Git is installed and available.

git config --global user.name "Your Name"
Set the author name used in new commits.

git config --global user.email "you@example.com"
Set the author email used in new commits.

git config --global init.defaultBranch main
Make main the initial branch of new repositories.

git init
Create a new local Git repository in the current folder.

git clone <url>
Copy a remote repository and configure origin.

git status -sb
Show the branch plus concise working-tree and staging state.

git diff
Inspect unstaged changes before selecting them.

git add <file>
Stage a named file for the next commit.

git add -p
Select individual change hunks to stage.

git diff --staged
Review exactly what the next commit will contain.

git commit -m "Message"
Create a local snapshot with an outcome-based message.

git commit --amend
Rewrite the most recent commit; only before it is shared.

git rm --cached <file>
Stop tracking a file that .gitignore cannot hide retroactively.

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Cheat Sheet 2 — Branch, Merge and Rebase

Isolate work and integrate it with the right strategy.

git branch
List local branches and identify the current branch.

git switch -c <branch>
Create and switch to a focused branch.

git switch main
Return to the main branch.

git switch -
Jump back to the previously checked-out branch.

git merge <branch>
Integrate a completed branch into the current branch.

git merge --no-ff <branch>
Force a merge commit so the branch remains visible.

git rebase main
Replay local commits on top of main; never on shared history.

git rebase --abort
Cancel an in-progress rebase and restore the prior state.

git cherry-pick <commit>
Copy one commit onto the current branch.

git branch -d <branch>
Delete a branch only after Git confirms it is merged.

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Cheat Sheet 3 — Sync, Remote and Release

Move history between the local repository and GitHub.

git remote -v
Inspect remote fetch and push endpoints.

git remote add upstream <url>
Register the source repository of a fork.

git fetch origin
Update remote-tracking references without changing working files.

git branch -vv
Inspect upstream tracking plus ahead/behind information.

git log HEAD..origin/main --oneline
List incoming commits before integrating them.

git pull --ff-only origin main
Fetch and fast-forward without creating an unexpected merge commit.

git push -u origin <branch>
Publish a branch and establish its upstream.

git push origin --tags
Publish annotated tags so releases are visible on GitHub.

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Cheat Sheet 4 — Undo and Rescue

Choose the safe undo for the state and sharing status.

git restore <file>
Discard an unstaged local change to a file.

git restore --staged <file>
Unstage a file while keeping its working-tree edits.

git reset --soft <commit>
Move HEAD only; keep the index and working tree.

git reset --mixed <commit>
Move HEAD and reset the index; keep the working tree.

git reset --hard <commit>
Move HEAD, index and working tree; discards work.

git revert <commit>
Create a new commit that safely reverses shared history.

git reflog
List where HEAD has been, so lost commits can be recovered.

git switch -c rescue <sha-from-reflog>
Recover a detached or reset commit onto a new branch.

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EXAM PREPARATION

From Course Completion to Certification

Use weighted practice data to plan your final revision

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CURRENT OFFICIAL DETAILS

Exam Details

Credential

GitHub Foundations · GH-900 · beginner level.

Fee

US$99 (regional pricing may apply)

Duration

100 minutes for the exam.

Delivery

Proctored online or at a Pearson VUE test centre.

Languages

English, Spanish, Portuguese (Brazil), Korean and Japanese.

Passing benchmark

Microsoft study guide states 700 or greater on its scoring scale.

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EXAM PREP

Four-Pass Preparation Strategy

1. Map your baseline by domain
2. Redo labs for weak domains
3. Use timed practice
4. Review explanations and retest

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TERTIARY INFOTECH PRACTICE PORTAL

Practice Exam

Practice mode

Learn from explanations and close knowledge gaps.

Exam mode

Simulate the 100-minute constraint.

Repeat

Retake after targeted revision, not immediately.

Open

https://exams.tertiaryinfotech.com/practice-exams/github/github-foundations

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SOURCE-ALIGNED

Official Preparation Resources

Study guide

Microsoft Learn GH-900 study guide and domain objectives.

Practice assessment

Official Microsoft practice assessment.

GitHub Learn

Certification resources and candidate guidance.

GitHub Education

Student preparation resources and eligible voucher offers.

Exam sandbox

Experience the exam interface before test day.

Your evidence

Use the fourteen Excel starter workbooks and completed examples as a personal revision record.

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PEARSON VUE

Exam-Day Checklist

Account

Use a personal Microsoft account so the credential remains portable.

Booking

Confirm the date, time zone, delivery mode and 100-minute appointment.

Identity

Prepare the accepted, unexpired identification named in the booking rules.

Environment

For online delivery, complete the system test and prepare a clear private workspace.

Check-in

Arrive early, follow the proctor instructions and protect exam content.

After the exam

Save the score report and use domain feedback to plan any follow-up revision.

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Course Feedback

Share

Tell us what worked and what to improve in the course feedback form.

Rate

Rate the trainer, the content and the labs honestly.

Certificate

Your certificate of completion is emailed after the course.

Support

Ask the trainer if you need help with any lab after class.

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Thank You

Collaborate clearly. Automate responsibly. Secure every change.

You are ready to apply GitHub foundations at work and prepare for GH-900.

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